CustomerSupport

Supplementary Example: Generative AI Chat Assistant

Identity-Sensitive i* Modeling of Brynjolfsson et al. [2023]

Case Introduction

This example is drawn from a large-scale deployment of a generative AI assistant at a Fortune 500 software company's customer support operation, studied by Brynjolfsson et al. [2023]. The company employed over 5,000 customer support agents handling technical issues via live chat. A GPT-based conversational AI assistant was introduced through a staggered rollout, providing agents with real-time suggestions during customer interactions—recommended responses, diagnostic steps, and resolution strategies. Agents could accept, modify, or ignore these suggestions at their discretion.

The deployment produced striking asymmetric outcomes: overall productivity increased by 14% (measured as issues resolved per hour), but this average masked a fundamental divergence. Novice agents improved by approximately 34%, while high-skill agents experienced minimal or slightly negative effects. The authors provide suggestive evidence that the AI "disseminates the best practices of more able workers and helps newer workers move down the experience curve"—in effect, the system captured expert behavioral patterns from historical data and redistributed them to less experienced agents.

In our models, **Jake** represents a veteran agent whose professional identity is grounded in rapid pattern recognition, high first-contact resolution rates, mentoring authority over junior colleagues, and a distinctive communication style developed through years of practice. **Mia** represents a novice agent whose baseline identity is still forming—she depends on scripts and escalation protocols, and her professional self-concept is aspirational rather than established. **Support Operations** is the organizational position responsible for throughput metrics, quality assurance, and workforce management.

Unlike the CleverLoan and NeuroYou cases, which are drawn from ethnographic or qualitative research with named individuals, the Brynjolfsson study is a large-scale quantitative analysis with anonymous agents. Jake and Mia are constructed as representative actors whose identity profiles are grounded in the study's empirical findings—specifically, its disaggregated productivity data, communication convergence patterns, and durable learning effects. The narration is transparent about this interpretive step.

Identity Tensions

The AI assistant generated identity effects that were invisible to the organization's productivity-oriented evaluation but analytically significant from an identity perspective:

- **For veteran agents (Jake):** The AI extracted Jake's accumulated expertise—his diagnostic patterns, resolution strategies, communication approaches—and redistributed them to the entire workforce without his participation, recognition, or consent. His skill differential, the foundation of his professional distinction, was systematically compressed. His informal mentoring role, through which he shaped junior agents' development, was displaced by the AI's automated dissemination of best practices. The organization measured a productivity gain; Jake experienced an erosion of the expertise-based identity that gave his work meaning.

- **For novice agents (Mia):** The same system accelerated her competence development dramatically. She gained access to expert-level resolution strategies immediately upon deployment, and—critically—retained measurable skill improvements even during periods when the AI was unavailable (durable learning). However, this rapid development was mediated entirely through AI dependency rather than through mentoring relationships with experienced colleagues. Her emerging professional identity was shaped by the AI rather than by the professional community.
- **For the customer relationship:** Customer sentiment improved and escalation rates decreased. But the communication patterns driving this improvement converged across all agents—the AI homogenized interaction styles by steering everyone toward the same optimal patterns. The distinctive relational qualities that individual agents brought to customer interactions were flattened.

This case is analytically distinctive because the identity effects were *unintended side effects* of a productivity-optimized deployment. Nobody designed for identity erosion; nobody mitigated it; the organization's evaluation framework—issues resolved per hour—was structurally incapable of surfacing it. The case thus demonstrates a different trajectory from the other three: **unintended homogenization**, where functional success coexists with unaddressed identity consequences.

‘As-Is’ SD Model

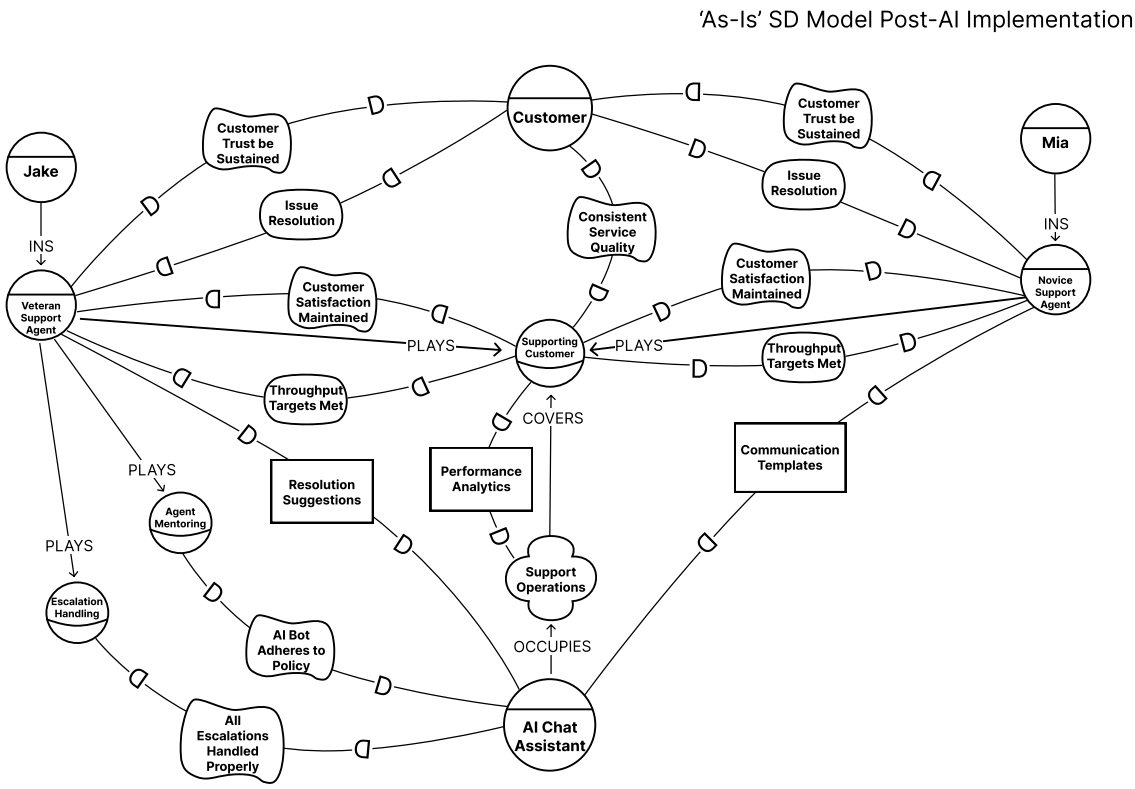

Figure 1: As-Is SD Model—Customer Support with AI Chat Assistant

What this model conveys

- **Agents, roles, and positions:** *Jake* and *Mia* are person-based agents who instantiate (INS) generic agents—*Veteran Support Agent* and *Novice Support Agent*, respectively. The *AI Chat Assistant* is a system agent. *Customer* is an external agent. The *Support Operations* position COVERS one formally recognized role:

  – **Supporting Customer**—the core operational role: receive chat, diagnose issue, resolve, close. Both the Veteran and Novice Support Agents PLAY this role. This is the only role the organization formally measures (issues resolved per hour), and where the AI’s augmentative effect is evaluated.

  Two additional roles appear in the model as *informal* role boundaries—they are **not** COVERS’d by Support Operations:

  – **Escalation Handling**—managing complex or escalated cases that exceed routine resolution capability (multi-system failures, high-frustration customers, edge-case diagnostics). Only the Veteran Support Agent (Jake) PLAYS this role; Mia does *not*. The Brynjolfsson findings document that the AI *reduces escalation rates*—improving customer sentiment and enabling novice agents to resolve cases that would previously have required escalation. This means the AI is compressing the volume of work that flows through Escalation Handling, silently eroding a role that was structurally distinctive to experienced agents like Jake.

- **Agent Mentoring**—guiding and socializing junior agents: modeling diagnostic reasoning, demonstrating communication style, explaining edge cases, providing feedback on handled chats. Jake's PLAYS link to this role remains structurally intact (shown as a solid line), but the role's function has been *functionally displaced*: Mia now receives expert-derived guidance from the AI Chat Assistant rather than from Jake. The displacement is invisible precisely because the structural link persists while the functional dependency has rerouted.

The informal status of Escalation Handling and Agent Mentoring is analytically significant: because neither role is formally COVERS'd by Support Operations, the organization has no metrics, no recognition mechanism, and no governance structure for these roles. The AI can erode them without triggering any organizational response—the only role the organization monitors is Supporting Customer, and that one is augmented, not eroded.

- **Role portfolio asymmetry—the structural key:** With three roles visible in the model, Jake and Mia are no longer structurally equivalent. Jake's professional identity is constituted across three roles (Supporting Customer + Escalation Handling + Agent Mentoring); Mia's involves only one (Supporting Customer). The AI erodes *two of Jake's three roles*—Escalation Handling through reduced escalation volume, Agent Mentoring through complete displacement—while augmenting both agents within the shared role. This asymmetry is the structural explanation for the Brynjolfsson study's central finding: why the same AI tool produces disproportionately positive effects for novices and minimal-to-negative effects for experts. Modeling only one role would make this invisible.

- **Core dependencies:** *Customer* depends on *Jake/Mia* for **Issue Resolution** (goal). Critically, the softgoal direction is *reversed*: *Jake* and *Mia* each depend on *Customer* for **Customer Trust be Sustained** (softgoal). This is an identity dependency, not a functional one—agents do not need customer trust to resolve tickets, but they need it to maintain their professional self-concept. Jake's version is grounded in years of earned expertise; customer trust validates his standing as a skilled practitioner. Mia's version is aspirational; customer trust signals that she is becoming competent. The AI's communication homogenization undermines both: when all agents sound alike, customers cannot distinguish Jake's expertise from Mia's AI-mediated responses, and Mia cannot be certain whether the trust she receives is directed at her or at the AI-generated communication she delivers. This reversed direction makes visible why communication convergence is an identity problem, not just a stylistic one.

  The model shows differentiated resource dependencies on the AI: *Mia* depends on *AI Chat Assistant* for **Communication Templates** (resource)—the AI provides phrasing, tone, and diagnostic scripts that shape her customer interactions. *Jake* depends on *AI Chat Assistant* for **Resolution Suggestions** (resource)—diagnostic and resolution recommendations that he can accept or override based on his judgment. This differentiation makes the homogenization mechanism precise: it operates primarily through the Communication Templates that shape Mia's professional voice, while Jake's relationship with the AI is mediated through suggestions he retains discretion over. *Support Operations* depends on both agents for **Throughput Targets Met** (goal) and **Customer Satisfaction Maintained** (softgoal). Within the informal *Escalation Handling* role boundary, *Jake* is the dependee for **Complex Case Resolution** (goal)—a dependency from *Support Operations* and from *Mia* (who escalates to Jake). The AI reduces the frequency of this dependency being activated, compressing Jake's distinctive contribution. Within the informal *Agent Mentoring* role boundary, *Mia* previously depended on *Jake* for **Guidance on Complex Cases** (resource) and **Professional Socialization** (softgoal). This dependency has been functionally

rerouted to the AI Chat Assistant—Mia's dependency on Communication Templates now serves the function that Jake's mentoring once served.

- **The invisible extraction:** The critical analytical feature of this model is the dependency that is *not* represented but structurally present: the AI Chat Assistant's effectiveness depends on historical behavioral data from high-skill agents like Jake. Expert knowledge flows *into* the AI (as training data) and *out* to novice agents (as suggestions), but this extraction has no corresponding dependency in the model. Jake is not a recognized *dependee* for the knowledge that powers the system. This one-directional flow—extraction without reciprocation—is structurally invisible in a conventional SD model, yet it is the mechanism through which expert identity is eroded.

- **Displaced mentoring as functional displacement:** The model shows Jake's PLAYS link to Agent Mentoring as a solid line—the role has not been formally reassigned. Yet upon analysis, the role's function has been displaced: Mia no longer seeks Jake's guidance because the AI provides it through Communication Templates. The displacement is invisible precisely because the structural link persists while the functional dependency has rerouted. This is analytically distinctive: unlike CleverLoan, where the destructive behavior was visible, or NeuroYou, where professional conflict surfaced tensions, here the mentoring displacement generates no organizational signal at all.

- **Homogenized communication through differentiated pathways:** The model shows that homogenization operates through different resource dependencies for each agent. Mia depends on the AI for **Communication Templates** that directly shape her professional voice—this is the primary homogenization channel. Jake depends on the AI for **Resolution Suggestions** that he retains discretion over, but the convergence documented in the Brynjolfsson study suggests that even partial adoption steers communication patterns toward uniformity. The differentiated resource profiles make the mechanism more precise than a blanket "everyone gets the same templates" reading: the AI shapes Mia's voice directly and Jake's voice indirectly, with the net effect being convergence across both agents' communication styles.

Key analytical point

The As-Is SD reveals a system that appears functionally healthy—all goal dependencies are satisfied, throughput targets are met, customer satisfaction is maintained—yet harbors four identity-level concerns—effects that, from a professional agency perspective, would be considered problematic even as the organization registers them as functional successes—invisible to conventional evaluation: (1) asymmetric knowledge extraction without recognition, (2) functional displacement of Agent Mentoring from Jake to the AI (while Jake's structural PLAYS link remains intact, making the displacement invisible), (3) compression of Escalation Handling volume that erodes expert-distinctive work, and (4) communication homogenization through differentiated AI resource dependencies. The three-role structure makes the asymmetric impact *structurally visible*: Jake's identity spans three roles and the AI erodes two of them; Mia's identity involves one role and the AI augments it. A single-role model would hide this entirely. The formal/informal role distinction deepens the diagnosis: the only role the organization formally recognizes and measures (Supporting Customer) is the one the AI augments; the two roles that constitute Jake's distinctive identity (Escalation Handling, Agent Mentoring) are informal and thus invisible to organizational metrics. The SD model's value here is diagnostic: it makes the *structure* of these concerns visible through role and dependency analysis, even though the organization's performance metrics register only success.

What this reveals beyond CleverLoan and NeuroYou

The CleverLoan As-Is model (LC-2) showed divergent identities within the same role facing a *substitutive* AI. The NeuroYou As-Is model (NY-2) showed bilateral identity threats across *different professions* collaborating on a shared system. The CustomerSupport As-Is SD shows a different pattern: an *augmentative* AI that produces asymmetric identity effects not through task substitution but through **role portfolio erosion and knowledge redistribution**. The formal/informal role distinction adds a structural explanation for why this erosion is invisible: the organization's COVERS relationship extends only to Supporting Customer—the role the AI augments—while the two identity-constitutive roles (Escalation Handling, Agent Mentoring) remain informal, uncovered, and unmeasured. The identity threat is not "your task is being automated" (CleverLoan) or "your professional paradigm is being challenged" (NeuroYou), but "your distinctive roles are shrinking while your routine role is augmented, and nobody notices because the metrics only measure the formal role."

Identity-Sensitive 'To-Be' SR Model

[Placeholder: CS-2 To-Be SR Model]

Replace this placeholder with the completed Strategic Rationale diagram (e.g., CS-2_ToBe_SR.pdf)

`\includegraphics[width=\textwidth]{CS-2_ToBe_SR}`

Figure 2: Identity-Sensitive To-Be SR Model—Customer Support Redesign

What this model conveys

This Strategic Rationale model presents an identity-sensitive redesign of the AI-assisted customer support system. It shows the internal reasoning of each actor in a reconfigured deployment that addresses the identity concerns surfaced in CS-1—not by removing the AI, but by restructuring how agents interact with it and how the organization recognizes the identity dimensions of AI-mediated work.

Role structure in the redesign

The three roles identified in CS-1 are preserved, but their organizational status and relationship to agents are fundamentally restructured. The redesign introduces a three-level **agent type hierarchy**: *Customer Support Agent* is the abstract agent type that OCCUPIES Support Operations. Two specializations exist: *Human Support Agent* (ISA Customer Support Agent) and *AI Support Agent* (ISA Customer Support Agent). Concrete agents are instances: Jake and Mia INS Human Support Agent; AI Chat Assistant INS AI Support Agent. Through ISA and INS inheritance, every concrete agent—human and AI—inherits the OCCUPIES relationship to Support Operations. All role access flows through the chain: INS → ISA → OCCUPIES → COVERS. No direct PLAYS links exist in the redesign.

This replaces the CS-1 structure, where agents reached roles through direct PLAYS links (via

INS intermediaries), and only Supporting Customer was COVERS'd by Support Operations. The redesign extends COVERS to all three roles:

- **Supporting Customer** remains the shared operational role, still augmented by the AI—productivity gains are preserved. All agents access it through the inheritance chain to Support Operations.
- **Escalation Handling** is **formalized** under Support Operations' COVERS scope. In CS-1, the AI was compressing this informal role's volume with no organizational signal. Now that the organization COVERS it, escalation activity is a formal metric—volume reduction becomes a visible signal, not a silent erosion.
- **Agent Mentoring** is **formalized and restored** under Support Operations' COVERS scope. In CS-1, Jake's direct PLAYS link remained structurally intact but was functionally displaced—Mia received guidance from the AI rather than from Jake. The redesign formalizes the role so that mentoring activity is organizationally measured and recognized, and restores its function through Jake's AI-augmented mentoring tasks.

The elimination of direct PLAYS links is the structural mechanism that prevents the identity erosion diagnosed in CS-1. When agents access roles through the INS → ISA → OCCUPIES → COVERS chain, the organization mediates that access: it can measure performance across all three roles, recognize contributions within each, and detect erosion if a role's function diminishes. The ISA split (Human vs. AI Support Agent) places both human and AI agents under the same organizational governance while preserving the type distinction that grounds identity analysis: human support agents carry person-based identity; AI support agents do not.

Jake's boundary—Expert Contributor

- Jake's central goal, *Customer Issues be Resolved* (J-G1), is achieved through a **means-ends OR decomposition** with two alternative tasks—a structure drawn from Horkoff and Yu's (2016) interactive goal model analysis. **Alternative A** (*Resolve via Passive AI Compliance*) represents the As-Is trajectory: accept all AI suggestions without exercising expert judgment. **Alternative B** (*Resolve Using Expert Judgment with AI Support*) represents the redesign choice: use AI suggestions as advisory input while applying Jake's own diagnostic expertise. Both alternatives achieve the functional goal—issues are resolved—but their contribution profiles to identity softgoals diverge sharply. Alternative A Hurts *Professional Expertise be Recognized* and Breaks *Mentoring Authority be Sustained*; Alternative B Makes *Professional Expertise be Recognized* and satisfies all three identity softgoals. A forward evaluation confirms that Alternative B is the identity-preserving choice—the one that achieves the same throughput while sustaining expert identity across all dimensions.
- **Under Alternative B**, Jake's tasks span all three roles: *Provide Real-Time Chat Support* (baseline function within Supporting Customer, shared with Mia—AI suggestions are now explanatory rather than opaque), *Handle Escalated Cases* (core competence within Escalation Handling—structurally reserved for high-skill agents, encompassing both receiving and resolving complex cases), *Mentor Through AI-Augmented Review* (restored Agent Mentoring function—Jake reviews AI-mediated interactions by junior agents, providing feedback informed by both his expertise and AI analytics), and *Validate AI Suggestions* (cross-role knowledge contribution—includes flagging incorrect recommendations and annotating edge cases with rationale, making Jake's expertise *visible and reciprocal*).

- **Identity-sustaining softgoals:** *Professional Expertise be Recognized* (++) receives positive contributions from *Validate AI Suggestions* and *Handle Escalated Cases*—encompassing both expert distinction and professional standing. *Mentoring Authority be Sustained* (+) receives a positive contribution from *Mentor Through AI-Augmented Review*—directly addressing the Agent Mentoring displacement identified in CS-1. *Knowledge Contribution be Attributed* (++) is a new softgoal that receives positive contribution from the organizational mechanism *Expert Knowledge Sources be Visible*—an organizational-level softgoal within the *Support Operations* boundary.

Mia's boundary—Developing Professional

- Like Jake, Mia's central goal *Customer Issues be Resolved* (M-G1) is achieved through a **means-ends OR decomposition**. **Alternative A** (*Follow AI Templates Uncritically*) represents the As-Is trajectory: accept AI communication templates without engaging with reasoning or developing personal style. **Alternative B** (*Engage Critically with AI Guidance*) represents the redesign choice: engage with AI reasoning, adapt suggestions to personal style, and develop independent judgment. Both alternatives resolve customer issues—and Alternative A may even be *faster* in the short term. But Alternative A Breaks *Communication Authenticity be Maintained* and Hurts *Autonomous Judgment be Cultivated*. Its one positive contribution is to *Professional Growth be Sustained* (Help)—reflecting the durable learning effect documented by Brynjolfsson et al., though narrowly scoped. Alternative B satisfies all three identity softgoals through a richer development pathway. A forward evaluation confirms Alternative B as the identity-preserving choice.

- **Under Alternative B**, Mia's tasks reflect a scaffolded development pathway rather than persistent AI dependency. In the As-Is state, Mia's activity was confined to Supporting Customer. In the redesign, Mia inherits organizational access to all three roles through the chain: Mia INS Human Support Agent ISA Customer Support Agent OCCUPIES Support Operations COVERS R1/R2/R3—giving her access to Escalation Handling as her competence develops. Her tasks: *Apply AI Suggestions with Understanding* (replacing passive acceptance with active comprehension—includes reviewing AI explanation before responding), *Resolve Independently When Confident* (progressive autonomy that activates as competence grows), *Develop Personal Resolution Patterns* (constructs her own professional style rather than replicating homogenized templates), and *Seek Mentoring on Complex Cases* (restores the relational dependency on Jake—reconnecting Mia to the Agent Mentoring role as a beneficiary of human guidance).

- **Identity-developing softgoals:** *Autonomous Judgment be Cultivated* (+) receives positive contributions from *Resolve Independently When Confident* and *Develop Personal Resolution Patterns*. *Professional Growth be Sustained* (++) receives contributions from both AI-mediated learning and mentoring—encompassing competence development and scope expansion as a single trajectory toward multi-role capability. *Communication Authenticity be Maintained* (+) receives positive contributions from *Develop Personal Resolution Patterns* and *Seek Mentoring on Complex Cases*—human mentoring transmits authentic professional style, not templates.

- **Progressive autonomy mechanism:** The AI Chat Assistant's task *Calibrate Support Level* adjusts the frequency and intrusiveness of suggestions based on demonstrated agent competence. Mia depends on the AI for **Scaffolded Learning Guidance** (resource) rather than **Resolution Suggestions**—a qualitatively different dependency that is designed to diminish over time rather than create permanent reliance. As Mia develops autonomous

competence, she moves toward PLAYING Escalation Handling—a role transition that the As-Is deployment's AI dependency would have permanently prevented.

AI Chat Assistant's boundary—Identity-Aware System

- The AI agent's internal rationale is redesigned to serve identity-sustaining functions alongside productivity.
- **Core tasks:** *Generate Explanatory Suggestions* (replacing opaque recommendations with reasoned guidance), *Calibrate Support Level* (progressive autonomy for developing agents), *Attribute Knowledge Sources* (making expert contributions visible in suggestions).
- **Softgoals:** *Expert Knowledge be Attributed* (+) ensures the AI does not erase the provenance of the expertise it disseminates. *Novice Autonomy be Developed* (+) ensures the AI's scaffolding function is oriented toward eventual independence.
- **Dependencies:** The AI now *depends on* Jake for **Validated Knowledge** and **Expert Annotations** (making the previously invisible extraction explicit and reciprocal), and *provides to* Mia **Scaffolded Learning Guidance** (not just raw suggestions).

Support Operations boundary—Identity-Sensitive Management

- The organizational position's internal rationale now includes identity-aware evaluation alongside throughput metrics.
- **Goals:** *Throughput Targets Met* (preserved from As-Is) AND *Professional Development Enabled* (new).
- **Tasks:** *Track Resolution Metrics* (operational baseline), *Monitor Skill Development Trajectories* (tracking agents' progression toward autonomous competence), *Recognize Knowledge Contributors* (organizational mechanism for expert attribution).
- **Softgoals:** *Expert Knowledge Sources be Visible* (++) is an organizational-level commitment that supports Jake's *Knowledge Contribution be Attributed*. *Communication Diversity be Valued* (+) counteracts the homogenization tendency by recognizing authentic communication styles alongside AI-optimized templates—supporting Mia's *Communication Authenticity be Maintained*.

Customer dependencies (external)

- *Customer* depends on agents for **Issue Resolution** (goal, preserved).
- The identity dependency from CS-1 is preserved: *Jake* and *Mia* each depend on *Customer* for **Customer Trust be Sustained** (softgoal). The redesign ensures this dependency can now be *satisfied*: because communication homogenization is counteracted and agents' distinctive expertise is visible, customer trust can differentiate between practitioners.

Agent Mentoring role boundary—means-ends alternatives

The Agent Mentoring role (R3) contains the most diagnostically revealing OR decomposition. Its goal, *Junior Agents be Professionally Developed* (R3-G1), has two alternative achievement paths: **Alternative A** (*Automate Guidance via AI*)—the AI provides all mentoring functions

without human involvement—and **Alternative B** (*AI-Augmented Human Mentoring*)—Jake mentors Mia with AI analytics as supporting input. Alternative A represents exactly what happened in the As-Is state: the AI displaced Jake's mentoring function. It Breaks *Professional Socialization be Sustained* (AI cannot transmit professional identity) and Breaks Jake's *Mentoring Authority be Sustained* (cross-boundary impact). Alternative B Satisfies professional socialization through human feedback and Makes Jake's mentoring authority by restoring and enriching his role. The goal evaluation retroactively diagnoses the As-Is state: CS-1 effectively chose Alternative A, which is precisely why professional socialization was severed and mentoring authority denied. The redesign makes this implicit choice explicit and selects the identity-preserving alternative.

Goal evaluation as design rationale

The three means-ends OR decompositions—Jake's J-G1, Mia's M-G1, and R3's R3-G1—collectively demonstrate why identity-sensitive modeling requires goal evaluation (Horkoff & Yu, 2016), not just dependency analysis. Each hard goal has alternative achievement paths that are *functionally equivalent* but *identity-divergent*. Throughput-oriented evaluation cannot distinguish between these alternatives because it measures only the hard goal. Goal evaluation, by propagating satisfaction labels through contribution links to identity softgoals, makes the identity cost of each alternative explicit and selectable. This adds a third analytical mechanism to the redesign: role formalization prevents *structural* erosion, reciprocal dependencies address *relational* extraction, and means-ends alternatives with goal evaluation address *decisional* identity preservation.

Key analytical point

The To-Be SR demonstrates that addressing the identity concerns surfaced in CS-1 does not require sacrificing the productivity gains that motivated the AI deployment. The redesign preserves the AI's augmentative function—Mia still benefits from expert-derived suggestions, the organization still achieves throughput improvements—while restructuring five critical dimensions:

1. **Role portfolio restoration and formalization through INS → ISA → OCCUPIES → COVERS**—Jake's three-role portfolio (Supporting Customer + Escalation Handling + Agent Mentoring) is preserved, the previously informal roles are brought under organizational COVERS scope, and all direct PLAYS links are replaced by an inheritance chain so that every agent—human and AI—accesses roles through the organizational position. The ISA split (Human vs. AI Support Agent) unifies governance while preserving the type distinction that grounds identity analysis.

2. **Knowledge extraction becomes reciprocal**—Jake is an explicit dependee for validated knowledge, not a passive source of training data.

3. **AI dependency becomes scaffolded**—Mia's interaction with the AI is designed to cultivate autonomous judgment and eventual multi-role capability rather than permanent single-role reliance.

4. **Communication homogenization is counterbalanced**—the organizational softgoal *Communication Diversity be Valued* and the agent-level softgoal *Communication Authenticity be Maintained* create structural pressure against the convergence tendency documented in the original study.

5. **Goal evaluation selects identity-preserving alternatives**—means-ends OR decompositions with contrasting contribution profiles make explicit the identity cost of design choices that throughput metrics treat as equivalent, providing a principled basis for selecting identity-sensitive implementations over functionally equivalent but identity-eroding ones.

The four requirement categories from the paper's Table 2 are all exercised:

- **Transparency:** The AI's *Generate Explanatory Suggestions* task provides reasoning alongside recommendations—met by design (cf. Santander success, CleverLoan failure).
- **Authority preservation:** Jake retains decision authority over complex cases and AI validation; Mia progressively acquires autonomous judgment rather than operating under permanent delegated authority.
- **Professional growth:** Both actors have development pathways—Jake through recognized knowledge contribution, Mia through scaffolded autonomy cultivation.
- **Expert recognition:** *Knowledge Contribution be Attributed* (++) and *Expert Knowledge Sources be Visible* (++) directly address the invisible extraction problem—the requirement category most distinctive to this case.

What this reveals beyond the other cases

The CustomerSupport To-Be SR tests the framework's ability to model a prescriptive redesign for a case where the *original deployment was measured as a success* by conventional metrics. In Santander, identity-sensitive design was already present (the TA manager championed it). In CleverLoan, the failure was evident (destructive behaviors, data manipulation). In NeuroYou, identity tensions surfaced through visible professional conflict. The CustomerSupport case is the most analytically demanding because the identity concerns are *invisible to the organization*—the system works, productivity rises, customer sentiment improves, and yet professional identities are being reshaped in ways that conventional evaluation cannot detect. The To-Be SR demonstrates that identity-sensitive modeling can surface requirements that no stakeholder has articulated, in a system that no stakeholder perceives as broken.

Coverage Analysis: What CustomerSupport Adds

Following the software testing analogy, the table below tracks which aspects of the identity-sensitive modeling framework are exercised by each case.

Modeling capability	**Santander**	**CleverLoan**	**NeuroYou**	**CustomerSupport**	**Notes**
Person-based identity (agent)	Tested (Tim)	Tested (Max, Anna)	Tested (Leo, Diana)	Tested (Jake, Mia)	All cases
Divergent identities in same role	—	Tested (Max vs. Anna)	—	Tested (Jake vs. Mia)	CL + CS
AI as substitutive threat	—	Tested	—	—	CL only
AI as augmentative tool	Tested	—	—	Tested	S + CS
Multi-role portfolio erosion	—	—	—	**Tested (3 roles → 1 effective)**	**New**
AI as knowledge extractor/ redistributor	—	—	—	**Tested**	**New**
Identity erosion through skill homogenization	—	—	—	**Tested**	**New**
Invisible mentoring displacement	—	—	—	**Tested**	**New**
Prescriptive redesign of a "successful" system	—	—	—	**Tested**	**New**
Bilateral identity threat (cross-profession)	—	—	Tested	—	NY only
Productive + destructive in same actor	—	Tested	—	—	CL only
Transparency as met/unmet requirement	Met (S)	Unmet (CL)	—	**Met by redesign**	S + CS
Authority preservation	Met (S)	Unmet (CL)	—	**Met by redesign**	S + CS
Expert recognition/ attribution	—	—	—	**Tested**	**New**
Scaffolded progressive autonomy	—	—	—	**Tested**	**New**
Communication authenticity vs. homogenization	—	—	—	**Tested**	**New**
Goal evaluation for identity-preserving design	—	—	—	**Tested**	**New**
Agent type hierarchy (INS→ISA→OCCUPIES→COVERS)	—	—	—	**Tested**	**New**

Modeling capability	Santander	CleverLoan	NeuroYou	CustomerSupport	Notes
Emergent mutual learning	—	—	Tested	—	NY only
Identity trajectory	Preservation	Destruction	Transformation	**Homogenization → Redesign**	Unique

Summary: CustomerSupport adds nine new test paths: (1) multi-role portfolio erosion—the AI compresses or displaces distinctive roles while augmenting the shared routine role, (2) AI as tacit-knowledge extractor without expert participation, (3) identity erosion through skill-level compression rather than task substitution, (4) role-level displacement of Agent Mentoring from human actor to AI, (5) prescriptive redesign of a deployment that conventional metrics judge as successful, (6) expert recognition and knowledge attribution as an identity requirement, (7) scaffolded progressive autonomy as an alternative to permanent AI dependency, (8) goal evaluation for identity-preserving design—means-ends OR decompositions with contrasting contribution profiles reveal the identity cost of functionally equivalent alternatives, and (9) agent type hierarchy (INS→ISA→OCCUPIES→COVERS)—a structured inheritance chain that unifies human and AI agents under the same organizational governance while preserving the type distinction that grounds identity analysis. Together with the other three cases, it completes a four-trajectory typology: designed preservation (Santander), unaddressed destruction (CleverLoan), emergent transformation (NeuroYou), and **unintended homogenization with identity-sensitive remedy** (CustomerSupport).

Domain-Specific Differences from Other Cases

Dimension	Santander	CleverLoan	NeuroYou	CustomerSupport
AI type	Augmentative (CV screening)	Substitutive (loan decisions)	Collaborative (ML hiring model)	Augmentative (chat suggestions)
Scale	Small team	Small unit	Two organizations	5,000+ agents
Source method	Qualitative	Qualitative	Ethnographic	Quantitative
Identity visibility	Surfaced by manager	Surfaced by resistance	Surfaced by conflict	**Invisible—surfaced only by modeling**
Organizational response	Designed mitigation	No mitigation	Negotiated adaptation	No identity-aware response
Our modeling role	Retrospective (success)	Diagnostic (failure)	Analytical (negotiation)	**Prescriptive (redesign)**
Temporal arc	4-year progression	Ongoing tension	Iterative mutual learning	Staggered rollout, ongoing